$n=5$
$p=9$

Decision tree

Date: __/__/20__

[1. Entropy]

$$-\frac{p}{p+n}\log_2\left(\frac{p}{p+n}\right)-\frac{n}{p+n}\log_2\left(\frac{n}{p+n}\right)$$

$$=\frac{-9}{9+5}\log_2\left(\frac{9}{9+5}\right)-\frac{5}{9+5}\log_2\left(\frac{5}{9+5}\right)$$

$$=\boxed{0.9403}$$

① Entropy of each attributes [Age group]

① $E(\text{Age group}=\text{Young})=-\frac{2}{5}\log_2\left(\frac{2}{5}\right)-\frac{3}{5}\log_2\left(\frac{3}{5}\right)$

$$=\boxed{0.971}$$

② $E(\text{Agegroup}=\text{middle})=$ $4=P$ $0=N$

$$=0$$

③ $E(\text{Agegroup}=\text{senior})=-\frac{3}{5}\log_2\left(\frac{3}{5}\right)-\frac{2}{5}\log_2\left(\frac{2}{5}\right)$

$$\boxed{=0.971}$$

[2. Average Info]

$$IG=0.94\left(\frac{5}{14}\times0.971+\frac{4}{14}\times0+\frac{5}{14}\times0.971\right)=\boxed{0.94}$$

$$0.94-(0.3475+0+0.3475)$$

$$\boxed{=0.245}$$

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② Entropy of Income level

(i) High (4 exp) P = 2 N = 2

$$\text{Entropy} = -\frac{2}{4}\log_2\left(\frac{2}{4}\right) - \frac{2}{4}\log_2\left(\frac{2}{4}\right)$$

$$= 0.5\log_2(0.5) - 0.5\log_2(0.5)$$

$$= -0.5(-1) - 0.5(-1) = \boxed{1}$$

(i) Medium P = 4 and n = 2

$$\text{Entropy} = -\frac{4}{6}\log_2\left(\frac{4}{6}\right) \frac{2}{6}\log_2\left(\frac{2}{6}\right)$$

$$= -2/3\log_2\left(\frac{2}{3}\right) - \frac{1}{3}\log_2(1/3)$$

$$= (0.38997) + (0.52832) = \boxed{0.918}$$

(i) low P = 3 n = 1

$$\text{Entropy} = -\frac{3}{4}\log_2\left(\frac{3}{4}\right) - \frac{1}{4}\log_2\left(\frac{1}{4}\right)$$

$$= 0.75\log_2(0.75) - 0.25\log_2(0.25)$$

$$= 0.31125 + 0.5 = \boxed{0.811}$$

Average Info Entropy

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$$\frac{4}{14}(1)+\frac{6}{14}(0.918)+\frac{4}{14}(0.811)$$

$$0.2857+0.3934+0.2317 = \boxed{0.9108}$$

$$\text{Gain} = 0.9403-0.9108$$
$$\boxed{= 0.0295}$$

③ Previous Purchase

(·) Yes (will buy) $P=6$ $n=1$

$$\text{Entropy} = -\frac{6}{7}\log_2\left(\frac{6}{7}\right)-\frac{1}{7}\log_2\left(\frac{1}{7}\right) = \boxed{0.592}$$

(·) No $P=3$ $N=4$

$$\text{Entropy} = -\frac{3}{7}\log_2\left(\frac{3}{7}\right)-\frac{4}{7}\log_2\left(\frac{4}{7}\right) = \boxed{0.985}$$

(·) Average Info Entropy

$$= \frac{7}{14}\times 0.592+\frac{7}{14}\times 0.985$$

$$= 0.296+0.4925 = \boxed{0.7885}$$

$$\text{(·) Gain} : 0.9403 - 0.7885 = \boxed{0.1518}$$

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Features	Average Info	Gain
Age group	0.695	0.245
Income level	0.9108	0.029
Previous purchase	0.7885	0.1518

Best feature : Age group

Final Decision tree